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Section - B

1 set question

Computer programming in C

chapter (11) pointers

Q What is pointer? JEE- 2013, 2014 What are pointer operation in c?

Pointer:- A pointer is a derived data type in C. A pointer is a variable that represents the location of the data item, such as a variable or an array element. pointers contain memory addresses as their values. pointers can be used to access and manipulate data stored in the memory.

Q. write down the advantages of pointer JEE- 2014

Advantages of pointer :-

pointers are used frequently in c. they include:

1. pointers are more efficient in handling arrays and data tables.
2. Pointers can be used to return multiple values from a function via function arguments.

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3. pointers permit references to functions.

4. the use of pointers arrays to character strings results in saving of data storage space in memory.

5. Pointers allow C to support dynamic memory management.

6. pointers provide an efficient tool for manipulating dynamic data structures such as structures, linked lists, queues, stacks and trees.

7. Pointers reduce length and complexity of programs.

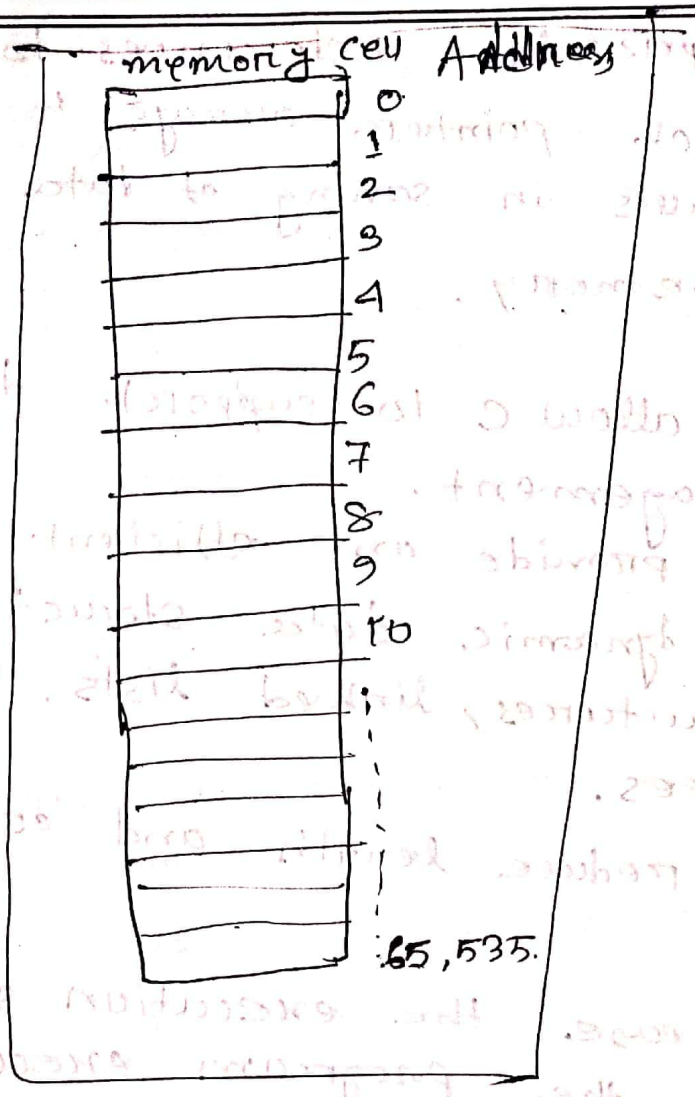
8. they increase the execution speed and thus reduce the program execution time.

Q. Understanding pointers

The computer's memory is a sequential collection of storage cells as shown in Fig-(1)

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Consider the following statement.

```
int quantity = 179;
```

This statement instructs the system to find a location for the integer variable quantity and puts the value 179 in that location. Assume the address location 5000 for quantity.

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Representation of variable as shown in fig-②

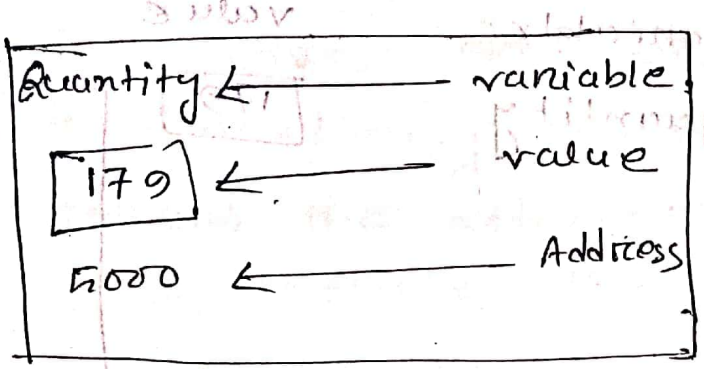


fig-② Representation of variable.

We may access to the value 179 by using either the name quantity or the address 5000. Variables that hold memory addresses are called pointer variables.

A pointer variable is a variable that contains an address, which is a location of another variable in memory.

Since a pointer is a variable, its value is also stored in the memory in another location. We assign the address of quantity to a variable P. The link between the variables P and quantity can be shown in fig-③. The address of P is 5048.

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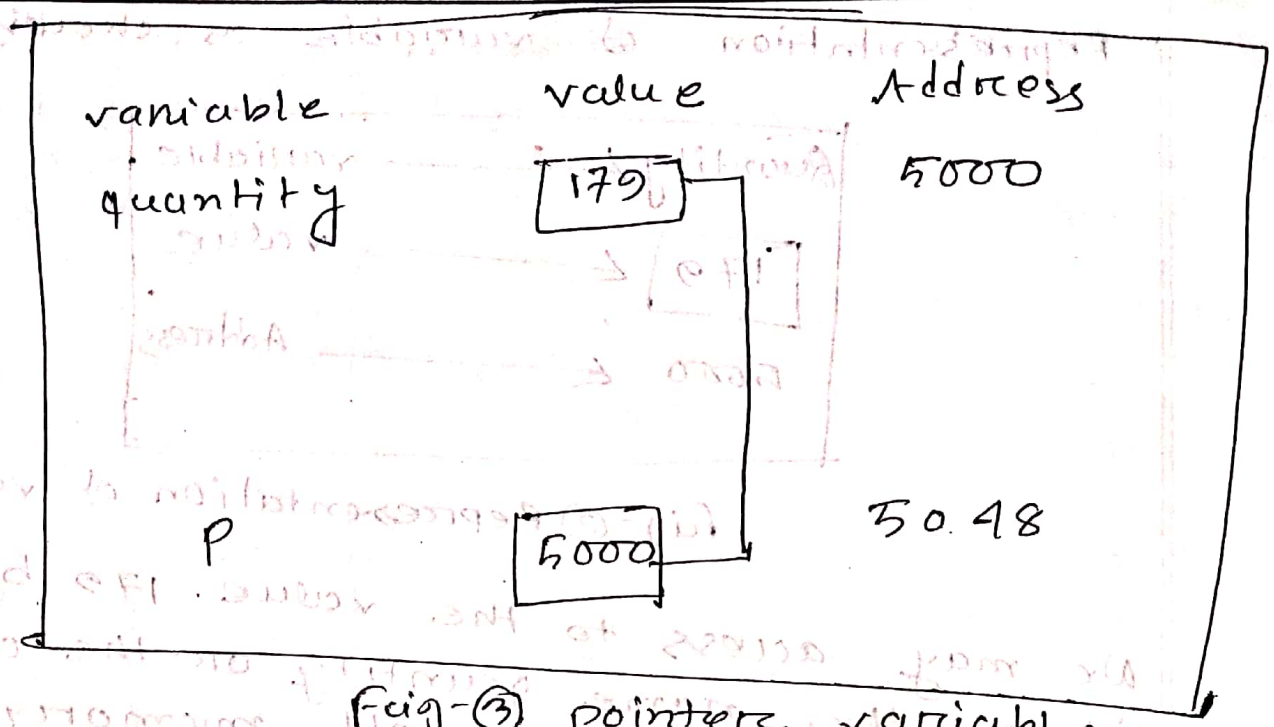


Fig-③ pointer variable

Since the value of the variable p is the address of the variable quantity we may access the value of quantity by using the value of p and p points to the variable quantity. This p ~~gets the~~ gets the name pointer.

How to access:

Q. Discuss ~~Accessing~~ the address of a variable using pointer.

Answer:- The actual location of a variable in the memory is system dependent and the address of a variable is not known.

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To determine the address of a variable we use the address operator &. The operator & immediately preceding a variable returns the address of the variable. for example, the statement.

```
p = &quantity;
```

would assign the address 5000 (the location of quantity) to the variable p. The & operator can be remembered as "address of".

Q. How to declare a pointer variable and initialize it? ② IEE-2013

Declaring pointer variables :- Pointer variables contains addresses that belong to a separate data type, they must be declared as pointers before we use them. The declaration of a pointer variable takes the following form:

```
data_type *pt_name;
```

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This tells the compiler three things about the variable pt_name;

1. The asterisk (*) tells that the variable pt_name is a pointer variable.
2. pt_name needs a memory location.
3. pt_name points to a variable of type data_type.

For example,

```
int *p;
```

declares the variable p as a pointer variable that points to an integer data type. The type int refers to the data type of the variable being pointed to by p and not the type of the value of the pointers.

Initialization of pointer variables :-

The process of assigning the address of a variable to a pointer variable is known as initialization. It is important to initialize

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pointers variables carefully before they are used in the program.

Once a pointers variables has been declared we can use the assignment operator to initialize the variable.

Example

```
int quantity;
int *p; /* declaration */
p = &quantity; /* initialization */
```

we can also combine the initialization with the declaration.

That is,

```
int *p = &quantity;
```

is allowed. The only requirement here is that the variable quantity must be declared before the initialization takes place. This is an initialization of p and not *p.

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The pointer variables always point to the corresponding type of data.

For example

```
float a, b;
int x, *p;
p = &a; /* wrong */
b = *p;
```

It is also possible to combine the declaration of data variable, the declaration of pointer variable and the initialization of the pointer variable in one step. For example,

```
int x, *p = &x;
```

It declares x as an integer variable and p as a pointer variable and then initializes p to the address of x.

We also define a pointer variable with an initial value of NULL or 0 (zero). The following statements are valued

```
int *p = NULL;
int *p = 0;
```

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Q. How is a variable accessed through its pointer. IEE-2013.

Answer :-

Accessing a variable through its pointer :-

Once a pointer has been assigned the address of a variable using unary operator * (asterisk) usually known as the indirection operator to access the value of the variable. Indirection operator also known as dereferencing operator. Consider the following statements.

```
int quantity, *p, n;
quantity = 179;
p = &quantity;
n = *p;
```

The first line declares quantity and n as integer variables and p as a pointer variable pointing to an integer.

The second line assigns the value 179 to quantity.

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The third line assigns the address of quantity to the pointer variable p.

The fourth line contains the indirection operator *.

When the operator * is placed before a pointer variable in an expression, the pointer returns the value of the variable of which the pointer value is the address. *p returns the value of the variable quantity, because p is the address of quantity. The * can be remembered as value address. Thus the value of n would be 179. The two statements

```
p = &quantity;
n = *p;
```

are equivalent to

```
n = quantity;
```

which in turn is equivalent to

```
n = quantity;
```

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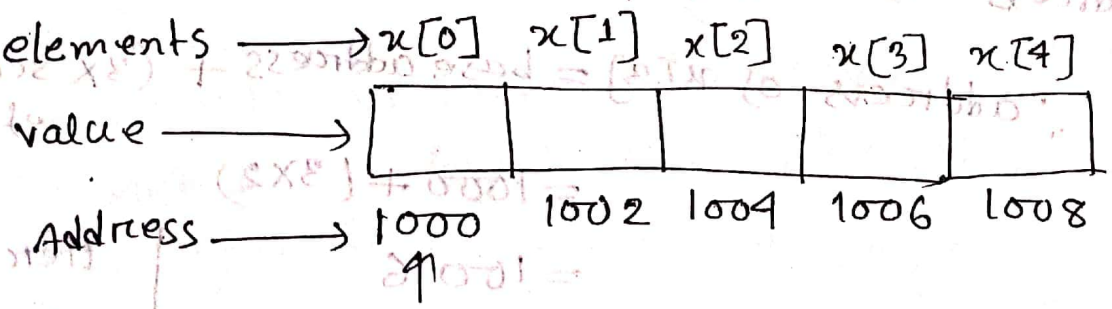
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Q.7 Explain how a pointer can be generated to an array. IEE-2011 or Explain the concept of pointer array with example. IEE-2015

Answer:- When an array is declared, the compiler allocates a base address and sufficient amount of storage to contain all the elements of the array in contiguous memory locations. The base address is the location of the first element of the array. Let we declare an array x as follows:

```
int x[5] = {1,2,3,4,5};
```

Suppose the base address of x is 1000. Hence we have



Suppose the base address of x is 1000, the location where x[0] is stored. That is

$x = \&x[0] = 1000;$

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If we declare p as an integer pointer, then the pointer p to point to the array x by the following statement

```
p=x;
```

This is equivalent to `p = &x[0];`

When handling arrays, instead of using array indexing, we can use pointers to access array elements.

Q. How calculate the address of an element.

Answer:- suppose we want to calculate the address of x[3].

∴ address of x[3] = base address + (3 x scale factor of int)

= 1000 + (3 x 2)

= 1006

Here, scale factor of int is = 2

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[Q.] Write a program using pointers to compute the sum of all elements stored in an array.

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program :-

```
main()
{
int *p, sum, i;
int x[5] = {5, 9, 6, 3, 7};
i = 0;
p = x;
printf("Element value address\n\n");
while (i<5)
{
printf("x[%od] %od %u\n", i, *p, p);
sum = sum + *p;
i++, p++;
}
printf("\n Sum = %od\n", sum);
printf("\n &x[0] = %u\n", &x[0]);
printf("\n p = %u\n", p);
}
```

Output :-

Element	value	Address
x[0]	5	166
x[1]	9	168
x[2]	6	170
x[3]	3	172
x[4]	7	174

sum = 30
&x[0] = 166
p = 176

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Q. Discuss about two-dimensional array representation of pointer.

Answer:-

pointers can be used to manipulate two-dimensional arrays. In a one-dimensional array x, the expression.

*(x+i) or *(p+i),

represents the element x[i]. Similarly an element in a two-dimensional array can be represented by the pointer expression as follows:

((a+i)+j) or *(*(p+i)+j)

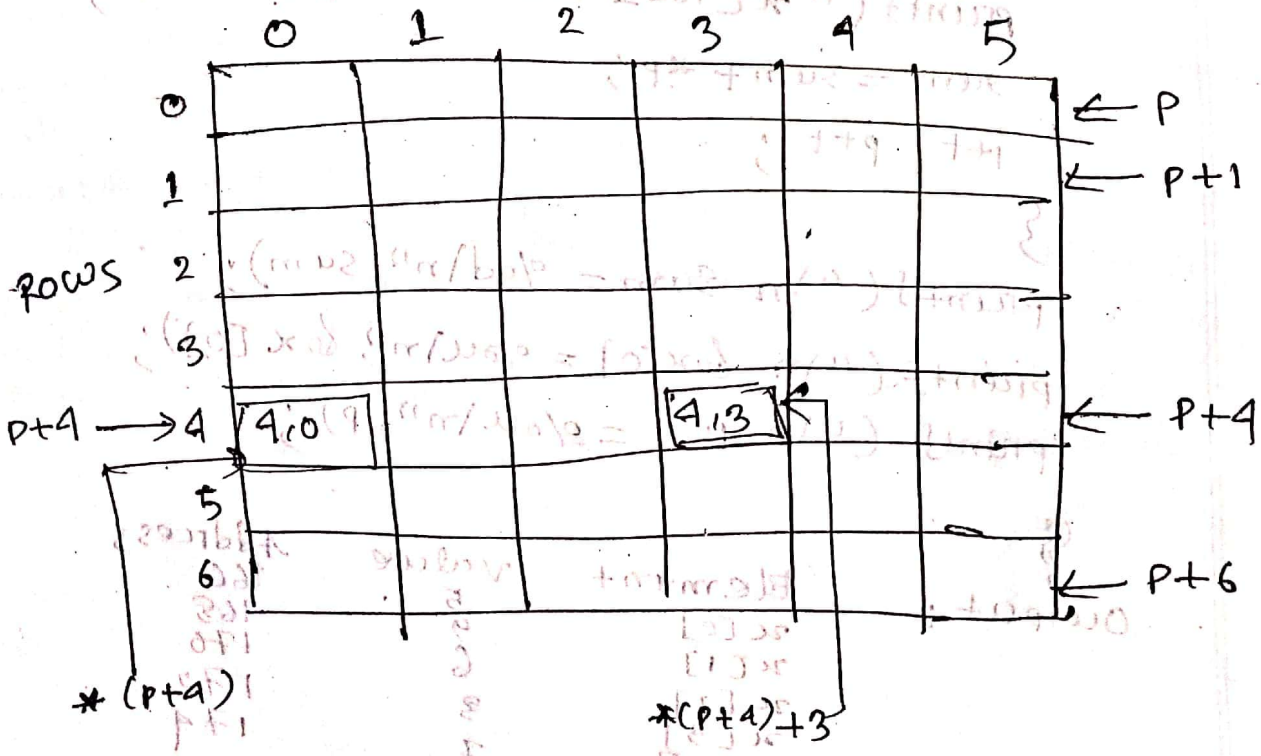


Fig: pointers to two-dimensional array.

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p ⟶ pointer to first row

pti ⟶ pointer to ith row

*(p+i) ⟶ pointer to first element in the ith row

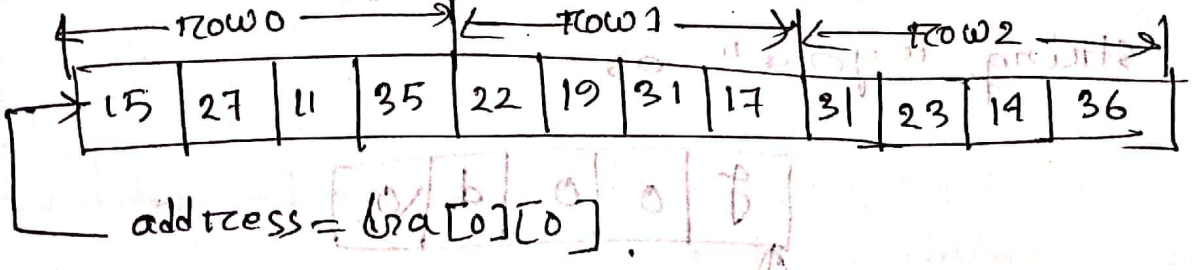
*(p+i)+j ⟶ pointer to jth element in the ith row

((p+i)+j) ⟶ value stored in the cell (i,j)

Suppose we declare an array as follows,

```
int a[3][4] = { {15, 27, 11, 35},
                {22, 19, 31, 17},
                {31, 23, 14, 36}
              };
```

The element of a stored as



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Q. Discuss pointers and character strings.

Answer:-

strings are treated like character arrays. they are declared and initialized as follows:

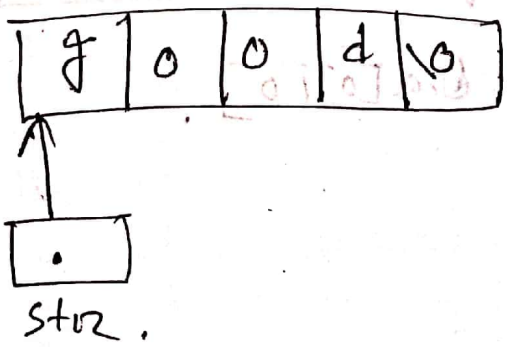
```
char str[5] = "good";
```

The compiler automatically inserts the null character '\0' at the end of the string.

To create strings using pointer variables of type char:

```
char *str = "good";
```

This creates a string and stores its address in the pointer variable str. The pointer str now points to the first character of the string "good" as:



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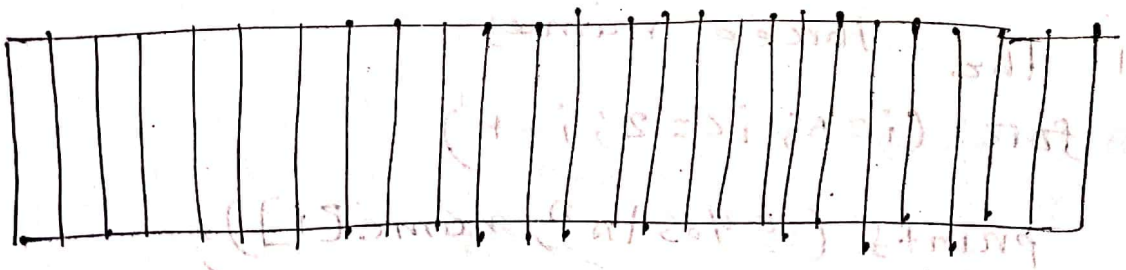
Q: write a program using pointers to determine the length of a character string.
Answer book page ~~(365)~~ new book article (11.5)

Q] Discuss about array of pointers.

Answer:- consider the following array of strings

```
char name[3][25];
```

This says that the name is a table containing three names, each with a maximum length of 25 characters (including null character). The total storage requirements for the name table are 75 bytes.



We can make a pointer to a string of varying length. For example

```
char *name[3] = {
                "New Zealand",
                "Australia",
                "India"
                };
```

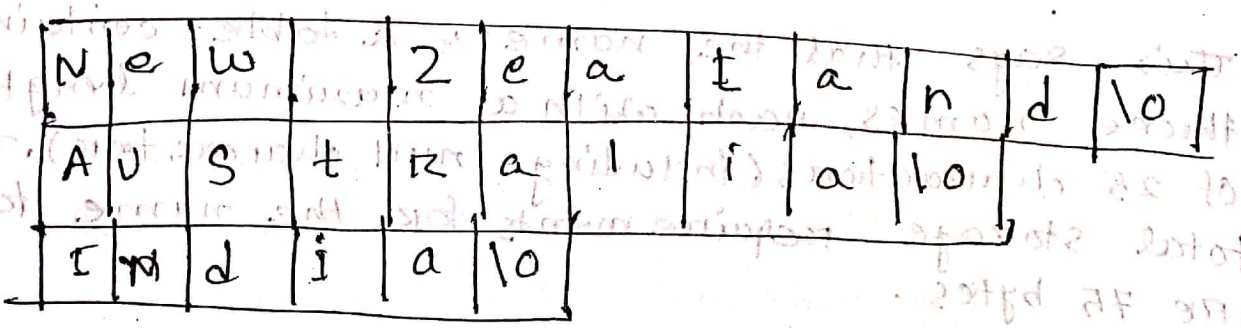
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declares name to be an array of three pointers to characters. each pointer pointing to a particular name as

name[0] ⟶ New Zealand
name[1] ⟶ Australia
name[2] ⟶ India.

The characters as shown



all the thriee names

```
for (i=0; i<=2; i++)
    printf ("%s\n", name[i]);
```

we many write into access the ith character in the ith name

*(name[i]+j).

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(b) Explain call-by-reference technique with an example. (3)

Call-by-reference:-

The process of calling a function using pointers to pass the addresses of variables is known as "call by reference".

The function which is called by 'reference' can change the value of the variable used in the call.

consider the following code.

```
main()
{
  int x;
  x = 20;
  change(&x);  /* call by reference or address */
  printf("%d\n", x);
}
change(int *p)
{
  *p = *p + 10;
}
```

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when the function change () is called, the address of the variable x, not its value, is passed into the function change (). Inside change (), the variable p is declared as a pointer and therefore p is the address of the variable x. The statement,

```
*p = *p + 10;
```

means 'add 10 to the value stored at the address p.' since p represents the address of x and the value of x is changed from 20 to 30.

call by reference provides a mechanism by which the function can change the stored values in the calling function. This mechanism is also known as "call by address" or "pass by pointers".

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Example : 11.6 [illegible] পৃষ্ঠা 228 ।

Q. Explain functions Returning pointers.

Answer : A function can return a single value by its name or return multiple values through pointer parameters. A function to return a pointer to the calling function. Consider the following code:

```
int *larger (int*, int*);
main()
{
  int a = 10;
  int b = 20;
  int *p;
  p = larger (&a, &b);
  printf ("%d", *p);
}
int *larger (int *x, int *y)
{
  if (*x > *y)
    return (x);
  else
    return (y);
}
```

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the function larger receives the addresses of the variables a and b, decides which one is larger using the pointers x and y and then returns the address of its location. And this case the address of b is returned and assigned to p and the output will be the value of b, namely 20.

Q. Explain pointers to functions.

Answer:-

A pointer to a function is declared as follows.

type (*fptr)();

This tells the compiler that fptr is a pointer to a function, which returns type value. A statement

type *gptr();

declare gptr as a function returning a pointer to type.

Q. Explain pointers and structures.

Answer:- Suppose product is an array variable of struct type. The name

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product represents the address of its zeroth element. Consider the following declaration:

```
struct inventory
{
 char name[30];
 int number;
 float price;
} product[2], *ptr;
```

This statement declares product as an array of two elements, each of the type struct inventory and ptr as a pointer. The assignment

```
ptr = product;
```

its member can be accessed using the following notation,

```
ptr → name
ptr → number
ptr → price.
```

The symbol → is called the arrow operator

The values of members of all the elements of product array,

```
for (ptr = product; ptr < product + 2; ptr++)
printf("\n%s %d %f\n", ptr → name, ptr → number, ptr → price);
```

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we also use
(*ptr).number,
to access the member number.

Q. What are the differences between call-by-value and call-by-reference ? (3)

Answer:- The differences are given below :

Call by reference	call by value
1. The process of calling a function using pointers to pass the address of variables is known as call by reference.	1. The process of passing the actual value of variable is known as call by value.
2. In call by reference the address of actual arguments is passed to formal arguments, hence any changed mode to formal arguments will also reflect in actual arguments.	2. In call by value a copy of actual arguments is passed to respective formal arguments.

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Call by reference	Call by value
3. In call by reference alterenation to actual arguments is possible within from called function ; therefore the code must handle arguments carefully else you get unexpected results.	3. In call by value, actual arguments will remain safe they cannot be modified accidently.

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